Algebraic stabilizations on anisotropic meshes: accuracy, nonlinear solvers and the DMP mesh conditions

galerkit

November 2026

Abstract

For convection–diffusion problems whose layers are known in advance, layer-adapted meshes such as Shishkin meshes use cells that are small across the layer and long along it. We study the eight algebraic stabilizations of galerkit (AFC with the BJK, Kuzmin and monotone convex limiters, MUAS, SMUAS, BBK, F-BBK, S-BBK) on uniform and Shishkin meshes with aspect ratios up to $6 \cdot 10^7$, for three problems with exponential and parabolic layers and $\varepsilon = 10^{-4}, 10^{-8}$: 638 runs. Shishkin meshes resolve the layers and reduce the L^2 error by up to four orders of magnitude for all methods. On these meshes, however, the Galerkin method is already stable, and the algebraic stabilization costs accuracy: for a problem with layers in both directions its errors are 5–40 times larger than those of Galerkin and SUPG, with first-order convergence away from the layers. The nonlinear problems become harder: the iteration numbers of the fixed point rhs grow with the anisotropy, and 62 of the 510 runs did not converge with any of three solvers. The DMP mesh conditions of the Kuzmin, MC and F-/S-BBK methods hold on Shishkin meshes split into two right triangles per cell and fail on the non-Delaunay crossed splitting; no DMP violation was observed even there. Switching the limiter off on the long edges along the layers cuts the iterations by factors 3–50, much more than replacing the Shishkin mesh by Bakhvalov-type or geometrically graded meshes (at most a factor 2).

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1 Questions

If the position of the layers is known, it makes sense to use mesh cells that are small in the direction of the convection (across an exponential layer) and large orthogonal to it; Shishkin meshes are the standard example. For the algebraic stabilizations we ask:

1. **Accuracy** of the solutions on such meshes, compared with uniform meshes and with Galerkin and SUPG.

2. **Efficiency**: the number of iterations needed to solve the nonlinear problem.

3. **DMP**: some methods satisfy the discrete maximum principle on all simplicial meshes, others only under a mesh condition. Are these conditions satisfied on anisotropic meshes, and is the DMP violated where they are not?

2 Set-up

Methods. P_1 elements, $\sum_j (a_{ij} + b_{ij}(u))u_j = g_i$ with the Galerkin matrix A and a symmetric stabilization matrix $B(u)$; the reaction term is not lumped.

method	galerkit	sufficient DMP condition (`dmp_certificate`)
AFC, BJK limiter	`afc`	none (any simplicial mesh)
MUAS, SMUAS	`muas`, `smuas`	none
BBK	`bbk`	none
AFC, Kuzmin limiter	`afc_kuzmin`	$\min\{a_{ij}, a_{ji}\} \le 0$ on every edge with a free node
AFC, monotone convex limiter	`afc_mc`	Delaunay type (Kuzmin condition checked as a proxy)
F-BBK, S-BBK	`fbbk`, `sbbk`	$\max\{a_{ij}, a_{ji}\} \le \tau_E = C_0\|E\|$

SUPG and the Galerkin method serve as references.

Problems (layers known), on $\Omega = (0,1)^2$:

- `exp_layer`: $\boldsymbol{b} = (1,0)$, $c = f = 0$, $u = (1 - e^{(x-1)/\varepsilon})/(1 - e^{-1/\varepsilon}) \in [0,1]$, an exponential layer at $x = 1$; with $f = c = 0$ the local DMP applies, so its violations can be counted node by node. $\varepsilon = 10^{-4}, 10^{-8}$.

- `boundary_layers`: $\boldsymbol{b} = (2,3)$, $c = 1$, exact solution with exponential layers at $x = 1$ and $y = 1$; $\varepsilon = 10^{-8}$.

- `parabolic_layers`: $\boldsymbol{b} = (1,0)$, $f = 1$, $u_b = 0$: exponential layer at $x = 1$, parabolic layers at $y = 0, 1$; no exact solution, but $0 \le u \le x \le 1$. $\varepsilon = 10^{-8}$.

Meshes. $n \times n$ cells, $n = 16, 32, 64, 128$: uniform ("/" diagonals) and Shishkin meshes with $n/2$ cells in each layer region of width $\tau = 2\varepsilon/\beta \ln n$ for an exponential layer (β the convection component across it) and $n/4$ cells in each region of width $\tau = 2\sqrt{\varepsilon}\ln n$ for the parabolic layers. Each Shishkin cell is split by "/", "\" or both diagonals (Figure 1). With two triangles per cell all triangles are right-angled and the mesh is Delaunay; the crossed splitting has angles up to 180° in the layer cells and is not Delaunay (Table 1).

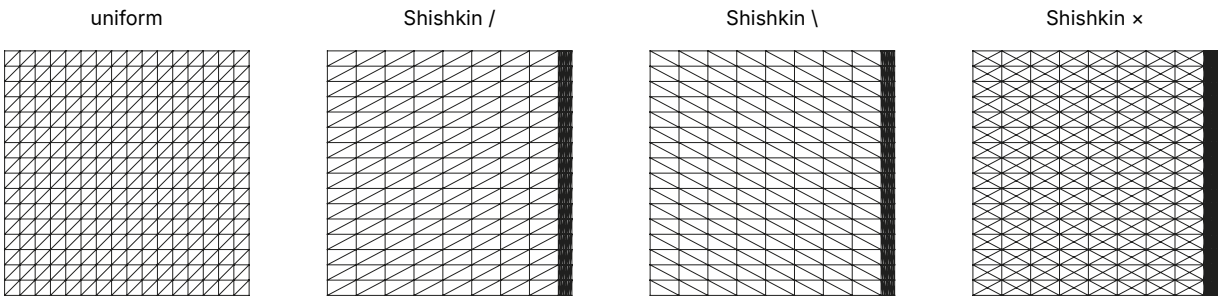


Figure 1: The mesh types for `parabolic_layers`, $n = 16$; drawn for $\varepsilon = 10^{-2}$ so that the layer cells are visible (at $\varepsilon = 10^{-8}$ the layer regions have widths of order 10^{-7} and 10^{-3}).

Table 1: The meshes at $n = 128$: nodes, largest aspect ratio (longest edge over its height), shortest edge, largest angle, and the edges with a positive off-diagonal entry of the P_1 Laplace matrix (not Delaunay).

case	mesh	nodes	aspect	$h_{\min}$	max angle	Laplace > 0
`exp_layer`, $\varepsilon = 10^{-4}$	uniform	16641	2.0e0	7.8e−3	90.0	0
`exp_layer`, $\varepsilon = 10^{-4}$	Shishkin /	16641	5.2e2	1.5e−5	90.0	0
`exp_layer`, $\varepsilon = 10^{-4}$	Shishkin \	16641	5.2e2	1.5e−5	90.0	0
`exp_layer`, $\varepsilon = 10^{-4}$	Shishkin ×	33025	1.0e3	1.5e−5	179.8	16320
`exp_layer`, $\varepsilon = 10^{-8}$	uniform	16641	2.0e0	7.8e−3	90.0	0
`exp_layer`, $\varepsilon = 10^{-8}$	Shishkin /	16641	5.2e6	1.5e−9	90.0	0
`exp_layer`, $\varepsilon = 10^{-8}$	Shishkin \	16641	5.2e6	1.5e−9	90.0	0
`exp_layer`, $\varepsilon = 10^{-8}$	Shishkin ×	33025	1.0e7	1.5e−9	180.0	16320
`boundary_layers`, $\varepsilon = 10^{-8}$	uniform	16641	2.0e0	7.8e−3	90.0	0
`boundary_layers`, $\varepsilon = 10^{-8}$	Shishkin /	16641	3.1e7	5.1e−10	90.0	0
`boundary_layers`, $\varepsilon = 10^{-8}$	Shishkin \	16641	3.1e7	5.1e−10	90.0	0
`boundary_layers`, $\varepsilon = 10^{-8}$	Shishkin ×	33025	6.2e7	4.6e−10	180.0	12224
`parabolic_layers`, $\varepsilon = 10^{-8}$	uniform	16641	2.0e0	7.8e−3	90.0	0
`parabolic_layers`, $\varepsilon = 10^{-8}$	Shishkin /	16641	1.0e7	1.5e−9	90.0	0
`parabolic_layers`, $\varepsilon = 10^{-8}$	Shishkin \	16641	1.0e7	1.5e−9	90.0	0
`parabolic_layers`, $\varepsilon = 10^{-8}$	Shishkin ×	33025	2.1e7	1.5e−9	180.0	16320

Solvers and measures. For every algebraic method: the fixed point rhs (`fp_rhs`), the standard solver for these schemes, with at most 3000 iterations (efficiency); if it does not converge (Euclidean residual $> 10^{-10}$), galerkit's Newton portfolio, and for $n \leq 32$ in addition the Anderson-accelerated fixed point with up to 20 000 iterations. Accuracy: the L^2 error, the maximal nodal error, and the maximal nodal error away from the layer regions. The energy norm is not used: on uniform meshes the quadrature does not see layers of width ε. DMP: the mesh condition of the method checked edge by edge, the local DMP for `exp_layer`, and the bounds of the exact solution.

3 Accuracy

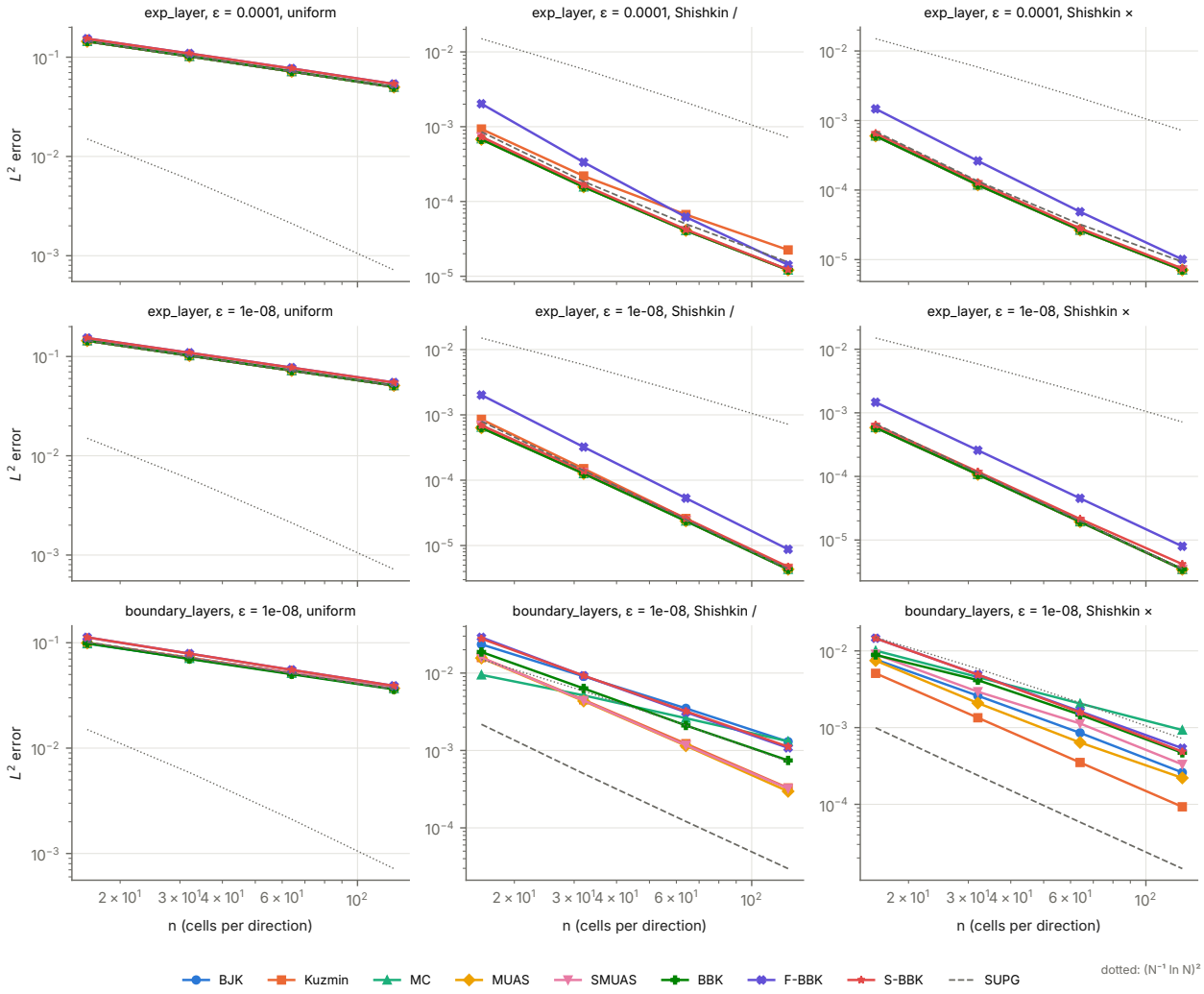


Figure 2: L^2 error against n on uniform, Shishkin "/" and crossed Shishkin meshes.

- **Shishkin meshes resolve the layers** (Figure 2). For `exp_layer` the L^2 error at $n = 128$ drops from $5 \cdot 10^{-2}$ on the uniform mesh to about $4 \cdot 10^{-6}$ ($\varepsilon = 10^{-8}$) for every method, with nearly the rate $(N^{-1} \ln N)^2$ and independently of ε. On the uniform mesh the nodal values of this layer, aligned with the mesh, are exact for most methods although the layer is not resolved; the maximal nodal error alone is therefore not a measure of quality here.

Table 2: `exp_layer`, $\varepsilon = 10^{-4}$, $n = 128$: maximal nodal error, its order in n (between $n = 64$ and 128), and the L^2 error.

	uniform			Shishkin /			Shishkin ×		
method	max	ord.	L^2	max	ord.	L^2	max	ord.	L^2
Galerkin	1.6e0	0.51	1.6e−1	7.8e−4	1.61	2.3e−5	9.4e−4	1.53	8.1e−6
SUPG	2.3e−14	-1.86	5.0e−2	1.6e−13	-2.56	1.5e−5	8.7e−4	1.45	9.3e−6
BJK	1.6e−14	-1.04	5.0e−2	7.7e−4	1.56	1.2e−5	9.4e−4	1.53	7.1e−6
Kuzmin	1.5e−14	-1.01	5.0e−2	3.0e−3	0.95	2.2e−5	9.4e−4	1.53	7.1e−6
MC	1.5e−14	-1.01	5.0e−2	7.8e−4	1.57	1.2e−5	9.4e−4	1.53	7.1e−6
MUAS	1.5e−14	-1.01	5.0e−2	7.7e−4	1.56	1.2e−5	9.4e−4	1.53	7.1e−6
SMUAS	1.5e−14	-1.01	5.0e−2	7.7e−4	1.56	1.2e−5	9.4e−4	1.53	7.1e−6
BBK	1.5e−14	-1.01	5.0e−2	7.7e−4	1.56	1.2e−5	9.4e−4	1.53	7.1e−6
F-BBK	1.2e−1	-0.01	5.4e−2	7.9e−4	1.51	1.4e−5	9.4e−4	1.53	1.0e−5
S-BBK	1.2e−1	-0.01	5.4e−2	7.8e−4	1.55	1.2e−5	9.4e−4	1.53	7.4e−6

Table 3: `exp_layer`, $\varepsilon = 10^{-8}$, $n = 128$: maximal nodal error, its order in n (between $n = 64$ and 128), and the L^2 error.

	uniform			Shishkin /			Shishkin ×		
method	max	ord.	L^2	max	ord.	L^2	max	ord.	L^2
Galerkin	2.4e3	1.44	6.1e2	8.4e−4	1.59	7.2e−5	7.2e−3	1.15	4.7e−4
SUPG	2.8e−14	-1.32	5.1e−2	7.2e−14	-1.65	4.4e−6	8.7e−4	1.45	3.5e−6
BJK	1.5e−14	-1.40	5.1e−2	7.7e−4	1.56	4.3e−6	9.4e−4	1.53	3.5e−6
Kuzmin	1.5e−14	-1.40	5.1e−2	3.0e−3	0.95	4.5e−6	9.4e−4	1.53	3.5e−6
MC	1.5e−14	-1.40	5.1e−2	7.8e−4	1.57	4.3e−6	9.4e−4	1.53	3.5e−6
MUAS	1.5e−14	-1.40	5.1e−2	7.7e−4	1.56	4.3e−6	9.4e−4	1.53	3.5e−6
SMUAS	1.5e−14	-1.40	5.1e−2	7.7e−4	1.56	4.3e−6	9.4e−4	1.53	3.5e−6
BBK	1.5e−14	-1.40	5.1e−2	7.7e−4	1.56	4.3e−6	9.4e−4	1.53	3.5e−6
F-BBK	1.2e−1	-0.00	5.5e−2	7.9e−4	1.51	8.7e−6	9.4e−4	1.53	8.0e−6
S-BBK	1.2e−1	-0.00	5.5e−2	7.8e−4	1.55	4.7e−6	9.4e−4	1.53	4.2e−6

Table 4: `boundary_layers`, $\varepsilon = 10^{-8}$, $n = 128$: maximal nodal error, its order in n (between $n = 64$ and 128), and the L^2 error.

	uniform			Shishkin /			Shishkin ×		
method	max	ord.	L^2	max	ord.	L^2	max	ord.	L^2
Galerkin	7.0e0	0.06	1.6e0	1.5e−2	0.78	6.5e−5	1.9e−3	1.34	3.3e−5
SUPG	2.9e−1	-0.04	3.6e−2	1.7e−2	0.89	3.0e−5	3.2e−3	1.30	1.5e−5
BJK	5.3e−2	0.82	3.6e−2	2.1e−2	0.96	1.3e−3	1.3e−2	0.90	2.6e−4
Kuzmin	1.6e−2	0.98	3.7e−2	4.5e−2	0.64	3.3e−4	3.1e−2	0.61	9.3e−5
MC	2.5e−2	0.94	3.7e−2	4.0e−2	0.59	1.3e−3	5.9e−2	0.56	9.3e−4
MUAS	1.6e−2	0.98	3.7e−2	2.4e−2	0.72	3.0e−4	6.5e−2	0.57	2.2e−4
SMUAS	1.6e−2	0.97	3.7e−2	3.2e−2	0.19	3.2e−4	2.9e−2	1.43	3.3e−4
BBK	6.8e−2	0.74	3.6e−2	3.0e−2	0.67	7.4e−4	3.2e−2	0.56	4.7e−4
F-BBK	1.7e−1	0.21	3.9e−2	2.9e−2	0.97	1.1e−3	1.8e−2	0.97	5.4e−4
S-BBK	1.7e−1	0.21	3.9e−2	3.7e−2	0.69	1.1e−3	2.9e−2	0.69	4.9e−4

- **The stabilization costs accuracy where it is not needed.** For `boundary_layers` on the "/" Shishkin mesh the Galerkin method (L^2 error $6.5 \cdot 10^{-5}$) and SUPG ($3 \cdot 10^{-5}$) converge with order 2. The algebraic methods are 5–40 times less accurate: Kuzmin and MUAS $3 \cdot 10^{-4}$ (order about 1.9; SMUAS the same, but from unconverged iterates), BJK and BBK 0.7–$1.3 \cdot 10^{-3}$ (order 1.5), MC $1.3 \cdot 10^{-3}$ (order 1). Away from the layers their maximal error, about 10^{-2}, converges only with order 0.7–1, against 1.6–1.8 for Galerkin and SUPG. On a mesh that resolves the layers, the Galerkin method is stable; the artificial diffusion of the limiters then only adds a first-order consistency error.

- **Splitting.** "/" and "\" give nearly the same errors. The crossed splitting doubles the number of nodes and gives the same accuracy for `exp_layer`; for `boundary_layers` it changes the ordering of the methods (Kuzmin, MUAS best).

- F-BBK is the least accurate method for `exp_layer` (about twice the error of the others).

4 Efficiency of the nonlinear solvers

Table 5: Iterations of the fixed point rhs at $n = 128$ (u: uniform, /: Shishkin /, ×: Shishkin ×); in colour: not converged within the limit (residual $> 10^{-10}$).

	`exp_layer`, $\varepsilon = 10^{-4}$			`exp_layer`, $\varepsilon = 10^{-8}$			`boundary_layers`, $\varepsilon = 10^{-8}$			`parabolic_layers`, $\varepsilon = 10^{-8}$		
method	u	/	×	u	/	×	u	/	×	u	/	×
BJK	0	997	3000	0	1075	3000	3000	3000	3000	3000	3000	3000
Kuzmin	0	95	355	0	95	355	176	964	3000	265	358	3000
MC	0	454	3000	0	461	3000	3000	3000	3000	3000	3000	3000
MUAS	0	449	3000	0	456	3000	176	3000	3000	291	1298	3000
SMUAS	0	406	3000	0	408	3000	176	3000	3000	252	3000	3000
BBK	0	1426	3000	0	1465	3000	3000	3000	3000	3000	3000	3000
F-BBK	246	3000	3000	251	3000	3000	3000	3000	3000	3000	3000	–
S-BBK	246	3000	3000	251	3000	3000	3000	3000	3000	3000	3000	–

- **The iteration numbers grow with the anisotropy.** Kuzmin, `exp_layer`, $\varepsilon = 10^{-8}$, $n = 128$: 0 iterations on the uniform mesh (the low-order start is already the solution), 95 on the Shishkin "/" mesh, 355 on the crossed one. MUAS, SMUAS and MC need 400–460 iterations on the "/" Shishkin mesh and do not converge within 3000 on the crossed one.

- **Robustness of fp_rhs** (converged runs out of 64 per method): Kuzmin 62, MUAS 56, MC 52, BBK 36, BJK 29, F-BBK 19, S-BBK 17. SMUAS converges on every uniform mesh but in only 22 of the 48 Shishkin runs.

- **Not converged at all.** With the Newton portfolio and the Anderson iteration as fallbacks, 62 of the 510 runs still end with a residual above 10^{-8}: BJK 25, F-BBK 13, S-BBK 12 (mostly `parabolic_layers`) and SMUAS 12 (all three Shishkin meshes of `boundary_layers`). The Anderson iteration converged in none of the 28 cases it was tried on. The errors reported for these runs are those of the best iterate.

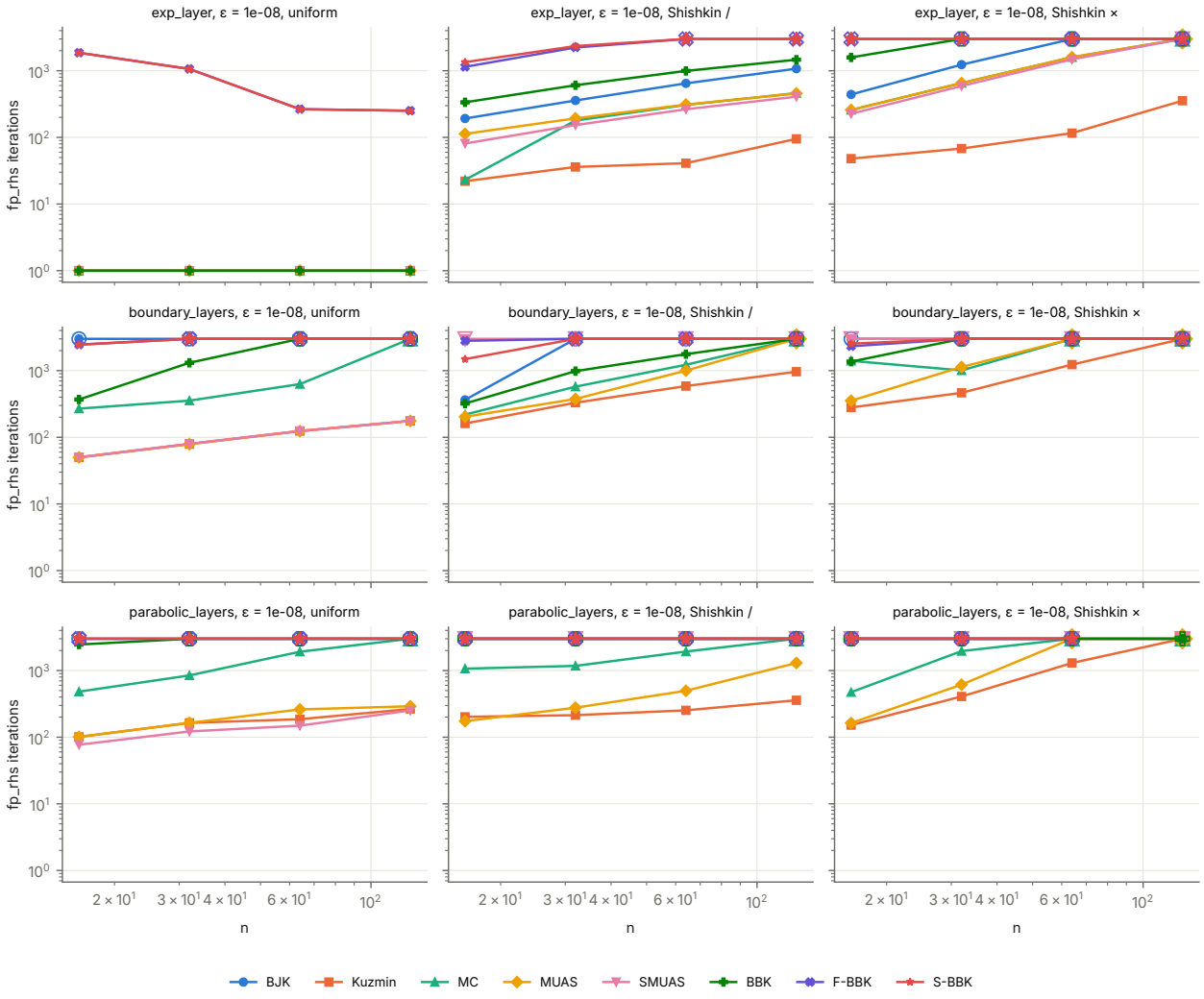


Figure 3: Iterations of the fixed point rhs; open circles: not converged within 3000 iterations.

5 The DMP and the mesh conditions

Table 6: DMP at $n = 128$: "ok" if the sufficient mesh condition of the method holds on every edge, otherwise the number of edges where it fails (colour); for `exp_layer` ($f = c = 0$) a second number gives the nodes violating the local DMP; †: the solution leaves the bounds of the exact solution by more than 10^{-8}.

	`exp_layer`, $\varepsilon = 10^{-4}$			`exp_layer`, $\varepsilon = 10^{-8}$			`boundary_layers`, $\varepsilon = 10^{-8}$			`parabolic_layers`, $\varepsilon = 10^{-8}$		
method	u	/	×	u	/	×	u	/	×	u	/	×
BJK	ok	ok	ok	ok	ok	ok	ok	ok	ok	ok†	ok†	ok†
Kuzmin	ok	ok	8192	ok	ok	8192	ok	ok	12224	ok	ok†	8192
MC	ok	ok	8192	ok	ok	8192	ok	ok	12224	ok	ok	8192
MUAS	ok	ok	ok	ok	ok	ok	ok	ok	ok	ok	ok†	ok
SMUAS	ok	ok	ok	ok	ok	ok	ok	ok	ok	ok	ok	ok
BBK	ok	ok	ok	ok	ok	ok	ok	ok	ok	ok	ok†	ok
F-BBK	ok	ok	8064	ok	ok	8064	ok	ok	12224	ok	ok†	–
S-BBK	ok	ok	8064	ok	ok	8064	ok	ok	12224	ok	ok†	–

- **Two right triangles per cell: the conditions hold.** On the uniform meshes and on the "/" and "\" Shishkin meshes the Kuzmin, MC and F-/S-BBK conditions hold on every edge, for all problems. The diffusion matrix of right triangles has no positive off-diagonal entries, and the non-lumped reaction term of `boundary_layers` ($c = 1$) does not break $\min\{a_{ij}, a_{ji}\} \le 0$, since convection dominates.
- **Crossed splitting: the conditions fail.** Where the layer cells are split into four triangles, the obtuse angles (up to 180°) make diffusion entries positive: the Kuzmin/MC condition fails on 128 to 12 224 edges, the F-/S-BBK condition at the larger n (8 064 edges for `exp_layer` at $n = 128$).
- **No observed DMP violation.** For `exp_layer`, where the local DMP applies, no method violates it on any mesh, including Kuzmin and MC on the crossed meshes where their condition fails. For `parabolic_layers` no method undershoots $u \ge 0$.
- **Overshoots of the exact bound.** On the "/" and "\" Shishkin meshes of `parabolic_layers`, BJK, Kuzmin, MUAS, BBK, F-BBK and S-BBK exceed the bound $u \le 1$ of the exact solution by up to 2–9 %; this is not a DMP violation ($f = 1$), but non-physical. MC and SMUAS stay below $1 + 6 \cdot 10^{-4}$, and no method overshoots on the crossed meshes.

6 Switching the stabilization off where the mesh resolves the layer

The switch. With the flag `afc_resolved_off=True` of `CDRSolver` (default `False`), every edge $E = \overline{x_i x_j}$ with mesh Péclet number

$$\mathrm{Pe}_E = \frac{|\boldsymbol{b}(x_E)|\,|x_j - x_i|}{2\varepsilon(x_E)} \le \mathrm{Pe}^* \qquad (\mathrm{Pe}^* = 1,\ \texttt{afc_pe_threshold})$$

gets $d_{ij} = b_{ij} = 0$: the scheme is the Galerkin method on the edges where the mesh resolves the solution. It acts on all eight methods through an edge mask of `AFCOperator`. The full speed $|\boldsymbol{b}|$ is used: with its component along the edge, $|\boldsymbol{b} \cdot (x_j - x_i)|$, every edge orthogonal to the flow would count as resolved, and the crosswind edges of the coarse cells (e.g. all vertical edges for $\boldsymbol{b} = (1, 0)$) would be switched off everywhere.

Already switched off there. The artificial diffusion of the algebraic methods,

$$d_{ij} = -\max\{a_{ij}, 0, a_{ji}\},$$

vanishes on every edge with $\max\{a_{ij}, a_{ji}\} \le 0$, which is the case on a diffusion-dominated edge of a right triangle. At $n = 128$ the meshes have between 8 128 and 32 640 edges with $\mathrm{Pe}_E \le 1$. For the problems `exp_layer` and `parabolic_layers`, none of them has $d_{ij} \neq 0$, so the switch changes nothing there. For `boundary_layers` it acts on about 4 000 edges at $n = 128$ where the reaction term ($c = 1$, not lumped) makes $\max\{a_{ij}, a_{ji}\} > 0$; switching them off loses the DMP guarantee on exactly these edges.

Table 7: `boundary_layers`, $\varepsilon = 10^{-8}$: L^2 error and fixed point rhs iterations with the stabilization everywhere / switched off on the edges with $|\boldsymbol{b}||E|/(2\varepsilon) \le 1$ (`afc_resolved_off`); *: final residual $> 10^{-8}$. Galerkin on the Shishkin "/" mesh, $n = 128$: 6.5e−5.

	Shishkin /				Shishkin ×			
	$n = 64$		$n = 128$		$n = 64$		$n = 128$	
method	L^2	it.	L^2	it.	L^2	it.	L^2	it.
BJK	3.5e−3 / 3.5e−3	3000 / 3000	1.3e−3* / 1.3e−3*	3000 / 3000	8.5e−4* / 8.5e−4*	3000 / 3000	2.6e−4* / 3.0e−4*	3000 / 3000
Kuzmin	1.2e−3 / 1.2e−3	587 / 587	3.3e−4 / 3.3e−4	964 / 964	3.5e−4 / 3.5e−4	1233 / 1233	9.3e−5 / 9.3e−5	3000 / 3000
MC	2.6e−3 / 2.6e−3	1225 / 1225	1.3e−3 / 1.3e−3	3000 / 3000	2.1e−3 / 2.1e−3	3000 / 3000	9.3e−4 / 9.3e−4	3000 / 3000
MUAS	1.2e−3 / 1.2e−3	1001 / 1001	3.0e−4 / 3.0e−4	3000 / 3000	6.4e−4 / 6.4e−4	3000 / 3000	2.2e−4 / 2.2e−4	3000 / 3000
SMUAS	1.2e−3* / 1.2e−3*	3000 / 3000	3.2e−4* / 3.2e−4*	3000 / 3000	1.1e−3* / 1.1e−3*	3000 / 3000	3.3e−4* / 4.0e−4*	3000 / 3000
BBK	2.1e−3 / 2.1e−3	1768 / 1767	7.4e−4 / 7.4e−4	3000 / 3000	1.5e−3 / 1.5e−3*	3000 / 3000	4.7e−4 / 4.7e−4*	3000 / 3000
F-BBK	3.1e−3 / 3.1e−3	3000 / 3000	1.1e−3 / 1.1e−3	3000 / 3000	1.6e−3 / 1.6e−3	3000 / 3000	5.4e−4 / 5.4e−4	3000 / 3000
S-BBK	3.2e−3 / 3.2e−3	3000 / 3000	1.1e−3 / 1.1e−3	3000 / 3000	1.6e−3 / 1.6e−3*	3000 / 3000	–	–

Result: it does not help (Table 7). For `boundary_layers` the L^2 errors and the iteration numbers are unchanged within a few per cent (Kuzmin, Shishkin "/", $n = 128$: $3.3 \cdot 10^{-4}$ in both cases, Galerkin $6.5 \cdot 10^{-5}$); BJK and SMUAS on the crossed mesh become slightly worse. The reason is where the error comes from: in the interior of the coarse region the algebraic methods are as accurate as the Galerkin method ($1.9 \cdot 10^{-3}$ against $1.5 \cdot 10^{-3}$ at $n = 64$); their large error ($2.3 \cdot 10^{-2}$) sits at the transition corner $(1 - \tau_x, 1 - \tau_y)$. It is caused by the long edges inside the layer strips (length $1/n$ along the layer, $\mathrm{Pe}_E \gg 1$ since $\boldsymbol{b} = (2, 3)$ has a component along them): there the limiters add diffusion along the layer, the Péclet criterion correctly regards these edges as unresolved, and the switch does not touch them. Reducing this error needs a criterion that sees the resolution of the solution, not only of the mesh (e.g. the limiter switched off on edges inside the layer region along the layer).

7 Switching the limiter off along the layers

The switch. With `afc_layer_off=True` (default `False`), $d_{ij} = b_{ij} = 0$ on the edges that run along a layer: edges whose adjacent cells all have aspect ratio $\mathrm{diam}(K)^2/(2|K|)$ at least 10 (the option `afc_aspect_threshold`), and on which the edge is a long edge ($|E| \ge \frac{1}{2}\mathrm{diam}(K)$) of every adjacent cell. These are exactly the edges named at the end of Section 6. The switch knows only the mesh; on a layer-adapted mesh the long edges inside the layer strips are where the solution varies slowly. `dmp_certificate` counts the switched-off edges with $\max\{a_{ij}, a_{ji}\} > 0$, where the DMP guarantee is lost.

Iterations. Median fixed point rhs iterations over the 16 runs per mesh and method (four cases, $n = 16$ to 128), counted over the runs that converge with and without the switch, Shishkin "/"

mesh: BJK 490 → 10, Kuzmin 128 → 26, MC 381 → 15, MUAS 308 → 17, SMUAS 209 → 7, BBK 983 → 35, S-BBK 2300 → 200; the number of converged runs does not decrease, except BJK (9 → 8) and F-BBK (5 → 4). On the crossed mesh the converged runs increase (BJK 4 → 8, MUAS 10 → 14, SMUAS 6 → 12, F-/S-BBK 1 → 9), except for BBK. Table 8 gives the numbers at $n = 128$: for `exp_layer` 5–10 iterations instead of 95–1465, for `boundary_layers` Kuzmin and MUAS need 128 instead of 964 and > 3000.

Accuracy. For `exp_layer` the errors are unchanged on the Shishkin mesh. For `boundary_layers` the error at the transition corner drops: Kuzmin, $n = 128$, maximal error away from the layers $1.4 \cdot 10^{-2} \to 2.5 \cdot 10^{-3}$, MUAS $6.9 \cdot 10^{-3} \to 2.7 \cdot 10^{-3}$, with the L^2 error $3.3 \to 2.9 \cdot 10^{-4}$ (Table 9). For `parabolic_layers` the overshoot of Kuzmin and MUAS grows slightly ($\max u_h = 1.008 \to 1.013$ at $n = 128$), while MC no longer stays below 1 but reaches $1 + 3 \cdot 10^{-4}$. No local DMP violation occurs for `exp_layer`.

8 Other layer-adapted meshes and the number of iterations

Question. Shishkin meshes are piecewise uniform: at the transition point the step size jumps from $O(\varepsilon \ln N/N)$ to $O(1/N)$. Do meshes with a smoother transition need fewer nonlinear iterations? We compare, with the same number of cells in the layer region ($n/2$ per direction, $\sigma = 2$, "/" diagonals; `galerkit.core.mesh.layer_points`):

- **Bakhvalov–Shishkin** (Linß): $x_i = -\sigma\varepsilon/\beta \, \ln\bigl(1 - (1 - N^{-1})\, i/(qN)\bigr)$ in the layer region, $q = \frac{1}{2}$, uniform outside; the step size grows smoothly up to the transition point;
- **Bakhvalov**: the mesh generating function $\varphi(t) = -\sigma\varepsilon/\beta \, \ln(1 - t/q)$ continued by its tangent through $(1, 1)$, a C^1 mesh generating function;
- **Durán–Lombardi**: $x_i = ih\varepsilon$ up to ε, then $x_{i+1} = (1 + h)x_i$, h chosen for N cells; geometric grading with ratio $1 + h$ and no transition point.

All three are anisotropic in the layer region as the Shishkin mesh; 480 runs, Tables 8 and 9.

Results without the switch.

- **Bakhvalov–Shishkin and Bakhvalov** reduce by about half the iterations of MC, MUAS and BBK for `exp_layer` and of Kuzmin, MC and MUAS for `boundary_layers` (at $n = 128$: Kuzmin 964 → 413, MUAS > 3000 → 592), and they are slightly more accurate there. Kuzmin on `exp_layer` needs more iterations (95 → 143), BJK no longer converges at all (0–1 of 16 runs, against 9 of 16), and SMUAS and F-/S-BBK remain unreliable. On the Bakhvalov mesh the algebraic methods are nodally exact away from the layer for `exp_layer` (L^2 error $6.6 \cdot 10^{-8}$).
- **Durán–Lombardi** gives the fewest iterations for `exp_layer` (Kuzmin 29 at $n = 128$) and lets SMUAS and BJK converge for `boundary_layers`. But at these N the grading $1 + h$ is large and its coarsest cells are much larger than $1/N$: for `boundary_layers` the L^2 error is 20 times larger than on the other meshes ($5.7 \cdot 10^{-3}$ for Kuzmin at $n = 128$), and for `parabolic_layers` Kuzmin and MUAS need three times more iterations (1175 against 358). It does not suit layers in two directions at moderate N.

Table 8: Iterations of the fixed point rhs at $n = 128$ on the layer-adapted meshes ("/" diagonals); "+ off": limiter switched off along the layers (`afc_layer_off`); in colour: not converged within 3000 iterations.

mesh	BJK	Kuzmin	MC	MUAS	SMUAS	BBK	F-BBK	S-BBK
`exp_layer`, $\varepsilon = 10^{-4}$								
uniform	0	0	0	0	0	0	246	246
Shishkin	997	95	454	449	406	1426	3000	3000
Bakhvalov–Shishkin	3000	143	226	236	241	736	3000	3000
Bakhvalov	3000	143	233	231	262	689	3000	2772
Durán–Lombardi	2693	55	214	253	250	255	3000	3000
Shishkin + off	6	7	6	9	6	5	3000	3000
Bakhvalov–Shishkin + off	440	9	9	9	9	7	3000	3000
Bakhvalov + off	145	9	9	8	6	26	3000	161
`exp_layer`, $\varepsilon = 10^{-8}$								
uniform	0	0	0	0	0	0	251	251
Shishkin	1075	95	461	456	408	1465	3000	3000
Bakhvalov–Shishkin	3000	143	227	238	241	748	3000	3000
Bakhvalov	3000	143	234	230	268	750	3000	2992
Durán–Lombardi	3000	29	102	148	140	221	3000	3000
Shishkin + off	7	7	7	10	6	6	3000	3000
Bakhvalov–Shishkin + off	50	8	10	8	8	7	344	176
Bakhvalov + off	3000	6	8	6	6	34	297	301
`boundary_layers`, $\varepsilon = 10^{-8}$								
uniform	3000	176	3000	176	176	3000	3000	3000
Shishkin	3000	964	3000	3000	3000	3000	3000	3000
Bakhvalov–Shishkin	3000	462	1623	643	3000	2241	3000	3000
Bakhvalov	3000	413	1537	592	3000	2275	3000	3000
Durán–Lombardi	618	669	1313	654	200	685	2972	2981
Shishkin + off	3000	128	1522	128	3000	2228	3000	3000
Bakhvalov–Shishkin + off	3000	128	1609	128	3000	2230	3000	3000
Bakhvalov + off	3000	128	1612	128	3000	2253	3000	3000
`parabolic_layers`, $\varepsilon = 10^{-8}$								
uniform	3000	265	3000	291	252	3000	3000	3000
Shishkin	3000	358	3000	1298	3000	3000	3000	3000
Bakhvalov–Shishkin	3000	325	3000	524	3000	3000	3000	3000
Bakhvalov	3000	363	3000	508	3000	3000	3000	3000
Durán–Lombardi	3000	1175	1006	1181	297	1407	3000	3000
Shishkin + off	3000	342	2335	343	3000	3000	3000	3000
Bakhvalov–Shishkin + off	3000	312	2261	316	3000	3000	3000	3000
Bakhvalov + off	3000	254	2654	328	3000	3000	3000	3000

Table 9: Accuracy on the layer-adapted meshes, $n = 128$, $\varepsilon = 10^{-8}$: L^2 error and maximal nodal error away from the layers for Galerkin, Kuzmin and MUAS; last column: $\max u_h$ of Kuzmin for `parabolic_layers` (exact bound 1).

	`exp_layer`						`boundary_layers`						parab.
	Galerkin		Kuzmin		MUAS		Galerkin		Kuzmin		MUAS		Kuzmin
mesh	L^2	away	L^2	away	L^2	away	L^2	away	L^2	away	L^2	away	$\max u_h$
uniform	6.1e2	2.4e3	5.1e−2	1.5e−14	5.1e−2	1.5e−14	1.6e0	7.0e0	3.7e−2	1.6e−2	3.7e−2	1.6e−2	0.998
Shishkin	7.2e−5	2.6e−4	4.5e−6	1.1e−5	4.3e−6	9.3e−6	6.5e−5	4.8e−4	3.3e−4	1.4e−2	3.0e−4	6.9e−3	1.008
Bakhvalov–Shishkin	2.5e−5	1.1e−4	3.0e−6	2.0e−5	2.6e−6	2.6e−5	8.5e−5	6.6e−4	2.9e−4	4.4e−3	2.9e−4	3.2e−3	1.013
Bakhvalov	1.7e−4	6.2e−4	6.6e−8	6.2e−15	5.3e−8	6.0e−15	1.5e−4	1.2e−3	3.0e−4	4.9e−3	2.9e−4	3.3e−3	1.015
Durán–Lombardi	1.5e−7	4.2e−5	3.1e−7	7.4e−6	1.5e−7	1.3e−5	4.0e−3	6.3e−3	5.7e−3	3.5e−2	5.7e−3	3.5e−2	1.000
Shishkin + off	–	–	4.3e−6	7.4e−6	4.3e−6	9.3e−6	–	–	2.9e−4	2.5e−3	2.9e−4	2.7e−3	1.013
Bakhvalov–Shishkin + off	–	–	1.5e−6	9.2e−5	1.5e−6	9.2e−5	–	–	2.9e−4	2.3e−3	2.9e−4	2.5e−3	1.012
Bakhvalov + off	–	–	2.3e−6	4.1e−5	2.3e−6	4.1e−5	–	–	2.9e−4	2.2e−3	2.9e−4	2.4e−3	1.011

Results with the switch. With `afc_layer_off` the mesh type hardly matters any more for Kuzmin, MC and MUAS: at $n = 128$, 6–10 iterations for `exp_layer`, 128 for `boundary_layers`, 250–340 for `parabolic_layers`, on all three meshes. The switch cuts the iterations by a factor 3–50, the choice of mesh by at most a factor 2. On the Bakhvalov meshes the switch costs accuracy for `exp_layer` (L^2 error $6.6 \cdot 10^{-8} \to 2.3 \cdot 10^{-6}$), since it replaces the nodally exact limiter by the Galerkin method on the graded cells; on the Shishkin mesh it costs nothing.

The best choice for few iterations. *The Shishkin mesh with the limiter switched off along the layers, with Kuzmin or MUAS*: the fewest iterations in all four cases, the smallest errors away from the layers for `boundary_layers`, and the simplest mesh. Bakhvalov–Shishkin with the switch is as good (and gives F-/S-BBK a chance), Bakhvalov without it is the best mesh for an aligned exponential layer.

What the mesh cannot fix. BJK, SMUAS, BBK, F-BBK and S-BBK fail for `boundary_layers` and `parabolic_layers` also on uniform meshes (Table 5); their failures are a property of the limiter and the fixed point iteration, not of the anisotropy. For these the sharper tools are on the solver side: damped and Anderson-accelerated fixed point iterations and Newton methods (compared for AFC by Jha and John), or differentiable (regularized) limiters that make Newton's method applicable (Badia and Bonilla). The switch along the layers remains useful with any of them, since it removes the nonlinearity from the edges where it is not needed.

9 Conclusions and next steps

1. On layer-adapted meshes all algebraic stabilizations resolve the layers, but they reduce the accuracy compared with the Galerkin method, which is stable there; the error concentrates at the transition corner. Switching the stabilization off on edges with mesh Péclet number ≤ 1 (`afc_resolved_off`) does not help (Section 6): the methods add no diffusion on those edges anyway, and the harmful diffusion acts on the long, unresolved edges along the layers.

2. The nonlinear problems become harder on anisotropic meshes. Kuzmin and MUAS remain the most robust; SMUAS, which is the most accurate method on general isotropic meshes, often fails to converge on Shishkin meshes.

3. The DMP conditions of the restricted methods are satisfied on Shishkin meshes split into right triangles and violated on crossed splittings, without observed DMP violations; the guaranteed methods need no condition. For anisotropic meshes the splitting into right triangles is therefore the safe choice.

4. Switching the limiter off on the long edges along the layers (`afc_layer_off`, Section 7) reduces the fixed point iterations by factors 3–50 and the error at the transition corner by up to a factor 5. Smoother layer-adapted meshes (Bakhvalov–Shishkin, Bakhvalov, Durán–Lombardi; Section 8) change the iterations by at most a factor 2 in either direction; with the switch the mesh type hardly matters. Fewest iterations: Shishkin mesh, switch on, Kuzmin or MUAS.

5. Open: anisotropic meshes not aligned with the axes, 3D, the solver failures of SMUAS on `boundary_layers`, and the role of σ in the transition point.

Reproducibility. galerkit branch `feature/anisotropic-afc`, folder

`examples/convection_diffusion/2d/anisotropic_afc`:

`run_aniso.py` (the runs), `make_report.py` (these tables and figures) and `results.json` (the 1688 runs: 638 on uniform and Shishkin meshes, 63 with `afc_resolved_off`, 507 with `afc_layer_off`, 480 on the graded meshes). F-BBK and S-BBK on the crossed Shishkin mesh of `parabolic_layers` at $n = 128$ were stopped after more than an hour and are missing.